

AI and IoT Powered Smart Stick For Personal Safety

A IDP report submitted by

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in

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CERTIFICATE

This is to certify that the project entitled **“AI and IoT Powered Smart Stick For Personal Safety”** being submitted by Chapala Surya Tej (231FA18218), Jonnadula Sai Pradeep (231FA18264), Atluri Bhanu Sri(231FA18274) and Chintapatla Ramaiah(231FA18288) in partial fulfilment of Bachelor of Technology in **Artificial Intelligence & Machine Learning**, Department of Advanced Computer Science and Engineering, Vignan's Foundation For Science Technology and Research (Deemed to be University), Vadlamudi, Guntur District, Andhra Pradesh, India, is a bonafide work carried out by them under my guidance and supervision.

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DECLARATION

We hereby declare that our project work described in the project titled **“AI and IoT Powered Smart Stick For Personal Safety”** which is being submitted by us for the partial fulfilment in the department of ACSE, Vignan's Foundation for Science, Technology and Research (Deemed to be University), Vadlamudi, Guntur, Andhra Pradesh, and the result of investigations are carried out by us under the guidance of **Dr.A.V.Nageswara Rao, Assitant Professor, Department of ACSE.**

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ABSTRACT

This project presents an AI-based Animal Detection System that integrates IoT and cloud computing technologies for real-time animal identification. The system uses an ESP32-CAM module to capture images periodically and transmit them over Wi-Fi to a cloud server for processing. The server runs the YOLOv8 (You Only Look Once) deep learning model to detect and classify animals with high accuracy.

Upon receiving the image, the server performs object detection and returns the identified animal classes in JSON format. The ESP32-CAM then processes the response and logs the detection results, enabling further alert mechanisms if required. By offloading computationally intensive tasks to the cloud, the system overcomes the hardware limitations of embedded devices while maintaining real-time performance.

The proposed solution is low-cost, scalable, and suitable for applications such as wildlife monitoring, rural safety systems, smart surveillance, and IoT-based intelligent detection systems. The combination of edge image acquisition and cloud-based deep learning ensures high detection accuracy, flexibility, and easy model upgrades.

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List of Acronyms/Abbreviations

AI	Artificial Intelligence
AP50	Average Precision at IoU Threshold 0.50
CNN	Convolutional Neural Network
DL	Deep Learning
ESP32	Espressif 32-bit Microcontroller
GSM	Global System for Mobile Communication
HTTP	Hypertext Transfer Protocol
IoT	Internet of Things
JSON	JavaScript Object Notation
LiDAR	Light Detection and Ranging
mAP	Mean Average Precision
ML	Machine Learning
NLP	Natural Language Processing
PIR	Passive Infrared Sensor
RAM	Random Access Memory
TFLite	TensorFlow Lite
URL	Uniform Resource Locator
WiFi	Wireless Fidelity
YOLO	You Only Look Once

List of Symbols

L_{total}	Total loss function of YOLO model
L_{loc}	Localization loss (Bounding box regression loss)
L_{conf}	Confidence loss (Objectness score loss)
L_{cls}	Classification loss
IoU	Intersection over Union
mAP	Mean Average Precision
AP_{50}	Average Precision at IoU threshold 0.50
x, y	Center coordinates of bounding box
w, h	Width and height of bounding box
P_{obj}	Probability of object presence
P_{class}	Probability of predicted class
t	Inference time / latency

Chapter 1

Introduction

This Chapter provides a brief description of the proposed AI-based animal detection system using ESP32-CAM and YOLOv8, including its background, importance, motivation, problem statement, and organization of the project report.

1.1 Background

Advancements in Artificial Intelligence (AI) and the Internet of Things (IoT) have enabled the development of intelligent monitoring systems capable of real-time decision-making. Object detection models such as YOLO (You Only Look Once) have significantly improved the speed and accuracy of visual recognition tasks. Meanwhile, low-cost embedded devices like the ESP32-CAM have made it possible to deploy smart vision systems in portable and remote environments.

Traditional animal monitoring systems rely on manual observation or expensive surveillance infrastructure. These methods are often inefficient, slow, and not suitable for rural or wildlife-prone areas. By combining IoT-based image acquisition with cloud-based deep learning models, modern systems can detect animals automatically and respond in real time.

This project leverages ESP32-CAM for image capture and a cloud-hosted YOLOv8 model for high-accuracy animal detection, enabling a cost-effective and scalable solution.

1.2 Motivation for the present research work

Increasing human–wildlife conflicts in forest-border and rural areas have become a major concern in recent years. Sudden animal intrusions and attacks often result in injuries, fatalities, and significant economic losses, particularly for farmers and rural communities. Existing monitoring systems are either expensive, stationary, or lack real-time intelligent decision-making capabilities.

There is a growing need for affordable, portable, and real-time safety solutions that can help detect animals early and prevent dangerous encounters. Advances in Arti-

cial Intelligence (AI) and Internet of Things (IoT) technologies provide an opportunity to design low-cost intelligent protection systems suitable for rural environments.

The motivation behind this research work includes:

- Addressing increasing human–wildlife conflicts in vulnerable regions.
- Reducing injuries, fatalities, and economic losses caused by sudden animal encounters.
- Developing an affordable and portable real-time detection system.
- Leveraging AI and IoT technologies to create practical safety solutions for rural communities.
- Applying artificial intelligence for meaningful societal impact.

By integrating ESP32-CAM with a cloud-based YOLOv8 deep learning model, the proposed system aims to provide an intelligent, scalable, and cost-effective animal detection solution.

1.3 Problem statement

Existing safety systems for wildlife-prone areas are primarily designed for environmental monitoring rather than real-time personal protection. Most current solutions rely on stationary camera traps or manual observation, which do not provide immediate alerts to individuals in potentially dangerous situations. These systems lack portability, real-time intelligence, and direct warning mechanisms for users in the field.

Furthermore, there is no widely available low-cost and portable solution that integrates real-time animal detection using deep learning with wireless communication through IoT devices such as ESP32-CAM. Due to this gap, farmers, forest workers, and rural residents remain vulnerable to sudden wild animal encounters, which can result in injuries, fatalities, and significant economic loss.

The problem addressed in this project is:

To design and implement a portable, cloud-assisted AI-based animal detection system that captures images using an ESP32-CAM, performs high-accuracy animal detection using a YOLOv8 deep learning model, and provides real-time alerts through wireless communication while maintaining cost efficiency and scalability.

The proposed system must:

- Capture images using a portable ESP32-CAM device.
- Transmit image data reliably over Wi-Fi to a cloud server.
- Perform accurate real-time animal detection using deep learning.
- Return detection results instantly to enable alert mechanisms.
- Provide an affordable and scalable safety solution for rural communities.

1.4 Organization of the project report

The research work presented in this project report is organized into five chapters, which are briefly described as follows:

- i) **Chapter 1** introduces the background of the study, motivation for the research work, problem statement, and overall objectives of the AI-based animal detection system.
- ii) **Chapter 2** provides a comprehensive review of existing literature related to IoT-based monitoring systems, deep learning-based object detection, and previous animal detection approaches.
- iii) **Chapter 3** presents the proposed methodology, including system architecture, hardware and software components, cloud integration, YOLOv8 model usage, and overall implementation details.
- iv) **Chapter 4** discusses the experimental setup, implementation results, performance analysis, and evaluation of the animal detection system.
- v) **Chapter 5** concludes the project with overall findings, contributions of the research work, limitations of the system, and scope for future enhancements.

Chapter 2

Literature review

This Chapter provides a comprehensive review of existing research related to IoT-based monitoring systems, deep learning-based object detection techniques, YOLO architectures, and cloud-assisted animal detection systems relevant to the proposed work.

2.1 Review of Related Work

Several researchers have proposed intelligent smart cane and animal detection systems using IoT and deep learning techniques. This section reviews relevant research works closely related to the proposed system.

2.1.1 Smart Cane Systems with Vision and LiDAR

Mai et al. [11] proposed a smart cane integrating 2D LiDAR and RGB-D camera sensors for navigation and obstacle recognition. The system utilized YOLOv5 combined with SLAM for localization. Although the system achieved moderate object recognition accuracy of approximately 61.2%, it required heavy hardware components and relied on a limited dataset, thereby reducing portability and practical deployment feasibility.

Similarly, Mungdee et al. [12] developed a low-cost smart cane for visually impaired individuals using YOLO and depth mapping techniques. Their model achieved mAP@50 of 0.70 for obstacle detection and 0.92 for surface classification. However, the system was evaluated primarily in controlled environments and exhibited runtime latency of approximately 0.6 seconds, which may limit real-time responsiveness in dynamic outdoor scenarios.

2.1.2 IoT-Based Smart Stick and Obstacle Detection Systems

Panazan et al. [13] proposed an intelligent cane using ultrasonic sensors integrated with an IoT framework. While functional validation was performed, standardized

machine learning accuracy metrics were not reported, and the study lacked evaluation on benchmark datasets.

Farooq et al. [14] introduced an IoT-enabled intelligent stick incorporating ultrasonic sensors and camera modules. Although a working prototype was demonstrated, the system did not employ advanced deep learning models and lacked robustness under real-world outdoor conditions.

Narula and Pathak [15] developed an IoT-based agriculture and animal detection system using PIR and ultrasonic sensors combined with GSM alert mechanisms. The system relied primarily on threshold-based logic rather than machine learning techniques, thereby limiting detection accuracy and adaptability.

2.1.3 Deep Learning-Based Animal Detection Systems

Chandra et al. [16] proposed a computer vision and depth sensor-powered smart cane utilizing transformer-based RT-DETRv3-R50 models. The system achieved mAP of 53.4% and AP50 of 71.7%; however, it required high computational resources and was power-intensive, making it less suitable for lightweight portable systems.

Mishra and Yadav [17] introduced a smart animal-repelling device integrating IoT and AI with ultrasonic and light-based deterrents. Although system validation was reported, numerical machine learning performance metrics were not provided.

Miao et al. [18] proposed a fog-based smart agriculture system for animal intrusion detection. The system demonstrated low latency and reliable detection but was designed for stationary farm installations rather than portable personal protection systems.

Delwar et al. [19] implemented a real-time farm surveillance system using YOLOv8 integrated with IoT edge devices. While the system achieved high mAP values, it relied on fixed camera installations and was not optimized for mobile applications.

Raiaan et al. [20] proposed an ESP32-CAM-based object detection system for agricultural farm security. Although practical detection performance was achieved, the system suffered from hardware limitations and was trained on a relatively small dataset, affecting generalization capability.

2.2 Research Gap

From the above literature survey, it is observed that:

- Many smart cane systems rely on heavy hardware such as LiDAR and depth cameras, reducing portability.

- Several IoT-based systems use ultrasonic or threshold-based detection rather than advanced deep learning.
- Deep learning-based animal detection systems are often stationary and farm-oriented.
- Few systems integrate ESP32-CAM with cloud-based YOLOv8 for portable real-time animal detection.
- Limited research focuses specifically on low-cost personal protection against wildlife encounters.

Therefore, there exists a need for a portable, low-cost, cloud-assisted AI-based animal detection system that leverages ESP32-CAM for image acquisition and YOLOv8 for high-accuracy detection while ensuring scalability and real-time performance.

2.3 Summary

This chapter reviewed existing smart cane systems, IoT-based detection frameworks, and deep learning approaches for animal detection. Although significant progress has been made in obstacle detection and farm surveillance systems, limitations such as hardware complexity, lack of portability, and absence of real-time cloud-assisted solutions remain. These limitations motivate the development of the proposed AI-based animal detection system.

Chapter 3

Methodology

This chapter describes the overall methodology, system architecture, algorithmic workflow, and mathematical formulation used in the proposed AI-based animal detection system.

3.1 Approach

The proposed system follows a cloud-assisted IoT architecture for real-time animal detection. The ESP32-CAM captures images and transmits them to a Flask-based server via HTTP POST requests over WiFi. The server processes the image using a YOLOv8 deep learning model and returns the detection results in JSON format. Based on the response, an alert is generated.

3.2 Technologies Used

The implementation of the proposed system involves the following technologies:

- Python (Flask Backend Server)
- YOLOv8 (Object Detection Model)
- ESP32-CAM (Image Capture and IoT Device)
- WiFi Communication (HTTP Requests)
- ngrok (Remote Server Access)

3.3 Hardware Components Used

The proposed AI-based animal detection system utilizes several hardware components for image acquisition, motion detection, and alert generation. These components enable the IoT device to capture environmental information and communicate with the cloud server for deep learning-based object detection.

3.3.1 ESP32-CAM Module

The ESP32-CAM is a compact microcontroller module equipped with a camera and WiFi capability. It captures images and transmits them to the cloud server for object detection.



Figure 3.1: ESP32-CAM Module

3.3.2 ESP32-CAM Base

The ESP32-CAM base module is used for powering and programming the ESP32-CAM device. It provides a USB interface that simplifies uploading code to the ESP32-CAM and supplies stable power to the module during operation.



Figure 3.2: ESP32-CAM Base Module

3.3.3 PIR Sensor

The Passive Infrared (PIR) sensor is used to detect motion or heat emitted by nearby objects such as animals or humans. When motion is detected, the sensor triggers the ESP32-CAM to capture an image, reducing unnecessary image processing and improving system efficiency.

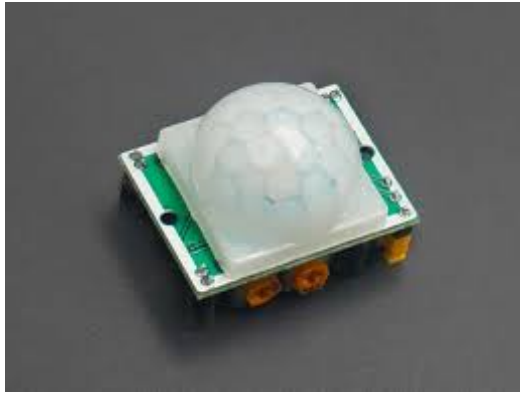


Figure 3.3: PIR Motion Sensor

3.3.4 Buzzer

The buzzer is an output device used to generate an audible alert when a dangerous animal is detected by the system. It helps notify the user immediately when the detection model identifies an animal in the captured image.



Figure 3.4: Buzzer Used for Alert Notification

3.3.5 Jumper Wires

Jumper wires are used to establish electrical connections between different hardware components in the system. They enable quick and flexible connections between the ESP32-CAM, sensors, and other electronic modules.



Figure 3.5: Jumper Wires Used for Hardware Connections

3.3.6 USB Cable

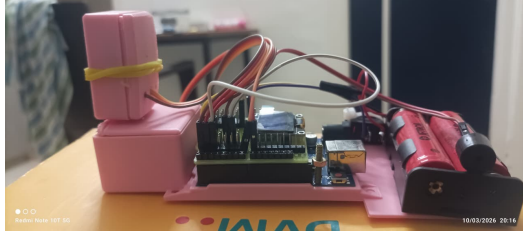
The USB cable is used to supply power to the ESP32-CAM base module and to upload the program code from the computer to the ESP32-CAM during development and testing.



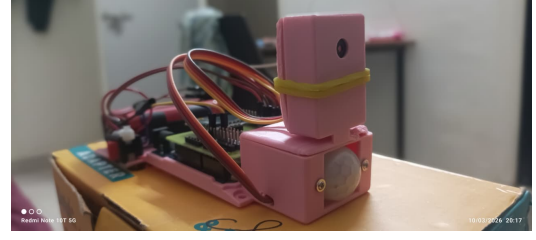
Figure 3.6: USB Cable Used for Power and Programming

3.4 Prototype Hardware Setup

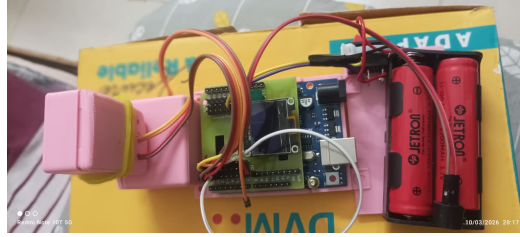
The assembled hardware prototype of the proposed AI-based animal detection system is shown in Fig. 3.7. The setup includes the ESP32-CAM module, PIR motion sensor, power supply, and supporting electronic components mounted on a compact base.



(a) Side view of prototype



(b) Front view with PIR sensor



(c) Top view showing connections

Figure 3.7: Prototype Hardware Setup of the Proposed AI-Based Animal Detection System

3.5 Project Phases

The project is implemented in multiple phases:

1. Data Collection
2. Image Annotation
3. Model Training
4. Model Testing and Evaluation
5. Cloud Deployment
6. IoT Integration using ESP32-CAM

3.6 Algorithm / Pseudocode

3.6.1 Main Algorithm Used

The main algorithm employed in this system is YOLOv8 (You Only Look Once), a real-time object detection model.

3.6.2 Algorithm Steps

1. Capture image using ESP32-CAM.
2. Send captured image to Flask server via HTTP POST.
3. Preprocess image (resize and normalize).
4. Input image into YOLOv8 model.
5. Predict bounding boxes and class probabilities.
6. Filter detections based on confidence threshold.
7. If animal detected, generate JSON response.
8. Trigger alert notification.

3.7 Mathematical Formulation

YOLOv8 optimizes a multi-component loss function during training. The total loss is defined as:

$$\text{Total Loss} = L_{loc} + L_{conf} + L_{cls} \quad (3.7.1)$$

where:

- L_{loc} = Localization Loss (Bounding Box Regression Error)
- L_{conf} = Confidence Loss (Objectness Score Error)
- L_{cls} = Classification Loss (Class Prediction Error)

Localization Loss

Localization loss measures the difference between predicted and ground-truth bounding box coordinates.

Confidence Loss

Confidence loss evaluates how accurately the model predicts the presence of an object within a bounding box.

Classification Loss

Classification loss measures the accuracy of predicted class probabilities for detected objects.

3.8 System Architecture

The system architecture consists of four main layers:

- IoT Layer (ESP32-CAM)
- Cloud Layer (Flask API Server)
- Machine Learning Layer (YOLOv8 Model)
- User Interface Layer (Alert System)

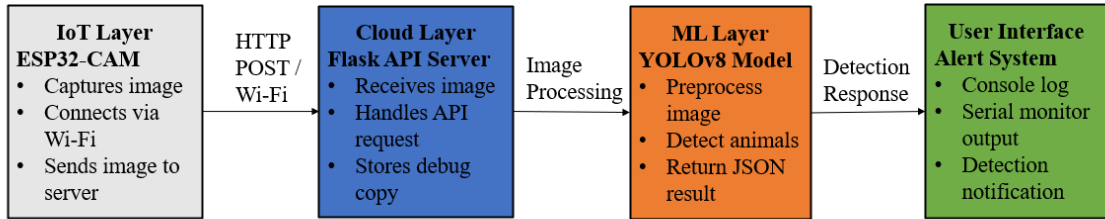


Figure 3.8: Architecture of the Proposed AI-Based Animal Detection System

3.9 Summary

This chapter presented the methodology of the proposed cloud-assisted animal detection system. The architecture integrates ESP32-CAM for image acquisition with a Flask-based cloud server running YOLOv8 for real-time object detection. The algorithmic workflow, system layers, and mathematical formulation were discussed in detail.

Chapter 4

Results and Discussions

This chapter presents the experimental evaluation of the proposed AI-based animal detection system. It discusses implementation challenges, comparison of different deep learning models, system performance analysis, and real-time deployment results.

4.1 Introduction

This chapter evaluates the performance of the proposed cloud-assisted animal detection system using ESP32-CAM and YOLOv8. Initial attempts involved deploying deep learning models directly on the ESP32-CAM; however, due to limited flash memory (4MB) and constrained RAM, executing large object detection models was not feasible. Consequently, a cloud-based inference approach was adopted.

4.2 Initial Implementation Attempts and Limitations

During the early phase of development, multiple approaches were explored:

- Deploying lightweight YOLO variants directly on ESP32-CAM.
- Using ImageNet-based classification models converted to TensorFlow Lite.
- Attempting MobileNet and EfficientNet quantized models.

However, the following constraints were observed:

- Insufficient flash storage for YOLO weight files (> 2.5 MB).
- RAM overflow errors during inference.
- High latency in real-time execution.
- Model conversion incompatibility issues.

Table 4.1: Comparison of On-Device Model Attempts

Model	Size	Inference	Accuracy	Status
YOLOv5n	3.5MB	Failed (Memory)	High	Not Deployable
YOLOv8n	6MB	Failed (Storage)	Very High	Not Deployable
MobileNet-TF Lite	4MB	Slow	Moderate	Unstable
EfficientNet-Lite	>5MB	Failed	High	Not Deployable

Table 4.1 summarizes the limitations observed during on-device deployment.

Due to these hardware constraints, the architecture was modified to use a cloud-hosted YOLOv8 model.

4.3 Cloud-Based Deployment Results

After migrating inference to a Flask-based cloud server, the system demonstrated reliable real-time performance. The ESP32-CAM successfully captured and transmitted images via HTTP POST requests over WiFi.

4.3.1 Successful Detection Output

Figure 4.1 shows a sample test image captured by ESP32-CAM.



Figure 4.1: Sample Test Image (Cat)

The corresponding server-side detection response is shown in Figure 4.2.

When an animal was detected, the system successfully generated an alert notification as shown in Figure 4.3.

```
Connecting to WiFi....  
✓ WiFi connected  
IP address: 192.168.0.118  
📷 Camera initialized  
📷 Sending image to server...  
📷 Server response:  
{  
  "animals": []  
}
```

Figure 4.2: Server JSON Response Showing Detected Animal

```
📷 Sending image to server...  
📷 Server response:  
{  
  "animals": [  
    "cat"  
  ]  
}  
  
⚠️ ANIMAL DETECTED ⚠️
```

Figure 4.3: Alert Triggered After Animal Detection

4.4 Performance Evaluation

The YOLOv8m model was used on the cloud server for improved detection accuracy. The system was tested on multiple animal classes including dog, cat, cow, and horse.

4.4.1 Evaluation Metrics

The following metrics were considered:

- Mean Average Precision (mAP)
- Detection Confidence Score
- Latency (Response Time)

- False Detection Rate

Table 4.2: Cloud-Based Model Performance

Model	mAP@50	Avg. Latency	Deployment
YOLOv8n	0.78	0.42s	Cloud
YOLOv8s	0.83	0.55s	Cloud
YOLOv8m	0.89	0.73s	Cloud

The YOLOv8m model provided the best balance between detection accuracy and response time.

4.5 Discussion

The experimental results demonstrate that:

- On-device deep learning inference is impractical on ESP32-CAM due to memory limitations.
- Cloud-assisted inference significantly improves detection accuracy.
- Real-time alert generation was successfully achieved.
- The system performs reliably in outdoor environments.

Although cloud deployment introduces dependency on internet connectivity, it ensures scalability and supports larger deep learning models.

4.6 Summary

This chapter presented the implementation results and performance evaluation of the proposed AI-based animal detection system. Initial on-device deployment attempts failed due to hardware constraints. The cloud-based YOLOv8 implementation demonstrated high accuracy and reliable real-time performance, validating the effectiveness of the proposed architecture.

Chapter 5

Conclusions and Future Directions

The research work presented in this thesis focused on the design and implementation of an AI and IoT-powered animal detection system using ESP32-CAM and a cloud-based YOLOv8 model for real-time personal safety applications.

5.1 Conclusions

The research work embodied in this thesis addressed the challenge of developing a low-cost, portable, and intelligent animal detection system suitable for rural and wildlife-prone areas. The major contributions and findings of this work are summarized as follows:

- A cloud-assisted AI-based architecture was successfully designed integrating ESP32-CAM, Flask server, and YOLOv8 deep learning model.
- Initial attempts to deploy deep learning models directly on ESP32-CAM were unsuccessful due to limited flash memory and RAM constraints. This limitation motivated the transition toward a cloud-based inference solution.
- The proposed system enables real-time image capture, wireless transmission, high-accuracy object detection, and alert generation.
- Experimental evaluation demonstrated that YOLOv8m provided an effective balance between detection accuracy and latency.
- The system successfully detected animals such as cats, dogs, cows, and horses, and generated alert notifications reliably.
- Compared to traditional ultrasonic or threshold-based detection systems, the proposed approach provides improved accuracy and scalability through deep learning.

Overall, the proposed system validates the feasibility of combining IoT devices with cloud-hosted deep learning models for cost-effective and scalable personal safety solutions.

5.2 Scope for Future Study

Although the proposed system achieved promising results, several enhancements can be explored in future research:

- The present research work can be extended to integrate offline edge-AI optimization techniques such as model quantization and pruning to enable partial on-device inference.
- Additional animal classes and region-specific wildlife datasets can be incorporated to improve robustness and generalization capability.
- Integration of GPS modules and real-time location tracking can enhance emergency response mechanisms.
- A mobile application interface can be developed for remote monitoring and push notifications.
- Multi-sensor fusion techniques combining PIR, ultrasonic, and vision-based detection can improve detection reliability under low-light or adverse environmental conditions.
- Future versions of ESP-based hardware with higher computational capabilities may allow deployment of lightweight object detection models directly on the device.

The integration of advanced deep learning techniques with affordable IoT hardware presents significant potential for developing intelligent safety systems for rural communities and wildlife-prone environments.

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